A COURSE-BASED PROJECT REPORT ON

CODE DUNGEON

A Gamified Java Programming Practice Platform

Submitted to the

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

in partial fulfilment of the requirements for

Object Oriented Programming Through JAVA Laboratory (25PC2CS201)

BACHELOR OF TECHNOLOGY
IN COMPUTER SCIENCE AND ENGINEERING

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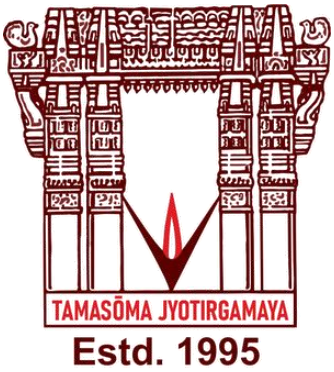
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OCTOBER 2026

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CERTIFICATE

This is to certify that the course-based project report entitled "Code Dungeon: A Gamified Java Programming Practice Platform" is a bonafide work carried out by the students listed below, in partial fulfilment of the requirements for the Automata and Compiler Design Laboratory (22PC2CS301) during the academic year 2026–2027.

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The work presented in this report is based on the project implementation and the project presentation supplied by the team.

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Estd. 1995

DECLARATION

We hereby declare that the course-based project work entitled "Code Dungeon: A Gamified Java Programming Practice Platform" is our own work carried out as part of the Automata and Compiler Design Laboratory. This report presents the system, database design, implementation flow and testing outcomes based on our project materials.

Place: Hyderabad

Date: October 2026

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ACKNOWLEDGEMENT

We express our sincere gratitude to the management and Principal of VNR Vignana Jyothi Institute of Engineering & Technology for providing the facilities required to complete this project.

We thank Dr. P. Anjusha, Assistant Professor, Department of CSE, for her guidance and support. We also thank Dr. V. Baby, Head of the Department of CSE, for the encouragement and academic support provided.

Finally, we thank our teammates and everyone who contributed to the successful completion of Code Dungeon.

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1. ABSTRACT

Code Dungeon is a desktop-based, gamified Java programming practice platform. Built using Java 21, Java Swing, JDBC and MySQL, it turns practice into short game challenges with MCQs, output-prediction questions, XP/points, enemies, level progression and a leaderboard. User accounts and learning progress are stored in a relational database.

2. INTRODUCTION AND OBJECTIVES

Beginners need a structured way to practise Java concepts and track improvement. Code Dungeon lets users sign in, choose an unlocked level, answer questions and earn points. Incorrect answers can be retried, and completing a level allows progression.

- Provide registration, login and logout for individual users.
- Organize practice into four levels: Java Basics; Conditions & Loops; Object-Oriented Programming; and Advanced Java.
- Support MCQs and code-output prediction, validate answers, and record attempts and scores.
- Show personal progress and a leaderboard to encourage continued practice.

3. SYSTEM DESIGN AND DATABASE

The Java Swing presentation layer displays forms, questions, progress and leaderboard screens. The application-logic layer manages authentication, question flow, answer checking, points and level unlocking. JDBC connects the Java application to MySQL.

Technologies used: Java 21, Java Swing, JDBC, MySQL, Maven, GitHub and MySQL Workbench.

Core database tables:

- users — account details and login-related information.
- levels — level names, order and progression/unlocking information.
- questions — question text, options, correct answers and level association.
- attempts — submitted answers, correctness and attempt history.
- scores — points/XP earned by users.
- progress — level completion and user progress.

The project uses INSERT to store records, SELECT with WHERE to retrieve user-specific data, UPDATE to change progress/scores, ORDER BY to rank leaderboard entries, GROUP BY with SUM to aggregate points, and JOIN to combine related records. JDBC PreparedStatement supports parameterized database operations.

4. IMPLEMENTATION AND TESTING

Main flow: registration/login → dashboard → choose an unlocked level → answer a question → validate the answer → update score and attempt history → display progress. Level completion unlocks the next level where applicable.

Core test outcomes reported in the supplied project presentation:

- Registration, login and logout flow — passed.
- MCQ and code-output prediction flow — passed.
- Correct/incorrect answer validation and retry behaviour — passed.
- Saving scores and attempts — passed.
- Level completion and next-level unlocking — passed.
- My Progress view and leaderboard — passed.

Separate screenshots or automated test logs were not included in the supplied project materials and are therefore not reproduced in this report.

5. CONCLUSION

Code Dungeon combines Java programming practice with game mechanics to make learning more engaging and measurable. The project demonstrates desktop UI development with Swing, database integration through JDBC, relational data management in MySQL and score/progress tracking. Future improvements could include more questions, hints and explanations, timed challenges, additional levels and richer progress analytics.

6. REFERENCES

- Code Dungeon project presentation supplied by the project team.
- Oracle Java documentation: https://docs.oracle.com/en/java/
- MySQL Reference Manual: https://dev.mysql.com/doc/
- Apache Maven documentation: https://maven.apache.org/guides/